

Reliable Tracking, Rugged Design, Open Interfaces



VT320 Vehicle Telematics Gateway

· LTE Cat-M

· GNSS

· CAN / OBD

· IP66

1. Product Overview

The Vehicle Telematics 320 (VT320) provides reliable operation in harsh environments even when the vehicle is powered off.

Key Features:

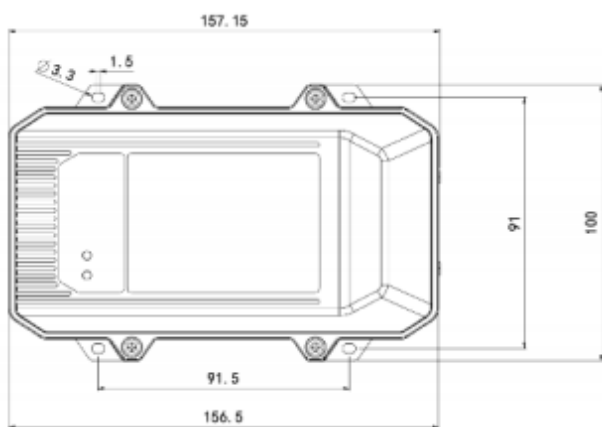
- **Reliable tracking:** GNSS multi-constellation positioning with inertial navigation support
- **Driving behavior monitoring:** gyroscope + inertial sensor monitoring for braking, acceleration, collision and other events
- **Extensive vehicle interfaces:** dual CAN bus, serial ports, I/O, Bluetooth, and 1-Wire
- **Data resilience:** cache supports continuous recording when Internet connectivity is unavailable
- **Vehicle-rugged design:** IP66 protection rating and wide working voltage range

Core Technical Specifications

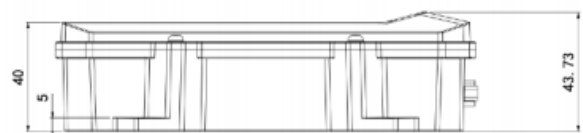
Metric	Spec
Cellular Network	LTE Cat.1 / LTE Cat-M1
Positioning	GPS / Galileo / Beidou / GLONASS + inertial navigation (DR)
Cloud Integration	AWS IoT, Azure IoT, Aliyun IoT, Wialon, Traccar, GPSWox, ThingsBoard
Network Features	TCP / UDP / HTTP / MQTT transport; SMS or FlexAPI reporting
Vehicle Data Access	OBD-II / J1939 / J1979 / J1708
Security	Certificate-based access capability; model-dependent operator certifications

Metric	Spec
Dimensions	Approximately 157.15 × 100 × 43.73 mm (6.19 × 3.94 × 1.72 in)
Vehicle Interfaces	2 × CAN, RS-232, RS-485, DI/DO, 1-Wire, analog input, dual-SIM
Power Supply	9-48 V DC
Operating Temperature	-40 °C ~ 85 °C (-40 °F ~ 185 °F)
Protection Rating	IP66
Battery	1200 mAh Ni-MH

2. Product Dimensions



Front View



Height Side View

Notes:

1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
2. All dimensions are approximate and **for reference only**.
3. Drawings **must not be used for manufacturing**.
4. Dimensions are subject to part and manufacturing tolerances.
5. Specifications may change without prior notice.

3. Hardware Specifications

Category / Parameter	Specification
Cellular & GNSS	
Network Type	LTE Cat.1 / Cat-M1
Antenna	LTE FPC built-in or external SMA (model dependent); GNSS ceramic built-in or external SMA
Satellite Support	GPS / Galileo / Beidou / GLONASS
Channels	72 channels
Initial Positioning Sensitivity	-167 dBm (initial positioning time 26 s)
Tracking Sensitivity	-157 dBm (hot start); -148 dBm (cold start)
Location Accuracy	2.5 m (CEP50)
Update Frequency	0.25 Hz ~ 10 MHz (configurable)
Bluetooth	Bluetooth 4.1
Inertial Sensor	
Acceleration Range	± 2 / ± 4 / ± 8 / ± 16 g
Angular Velocity Range	± 125 / ± 250 / ± 500 / ± 1000 / ± 2000 dps
Inertial Navigation	Supports dead-reckoning (DR)
Working Conditions	
Working Voltage	9–48 V DC

Category / Parameter	Specification
Operating Temperature	-40 °C ~ 85 °C (-40 °F ~ 185 °F)
Humidity	95% RH @ 50 °C (non-condensing)
ESD	IEC 61000-4-2 (4 kV test)
Protection Rating	IP66
Battery Capacity	1200 mAh
Battery Material	Ni-MH
Battery Temperature Range	Charging: 0 °C ~ 45 °C (32 °F ~ 113 °F); Discharging: -40 °C ~ 85 °C (-40 °F ~ 185 °F)
Storage Temperature	For 1 month: -40 °C ~ 55 °C (-40 °F ~ 131 °F); For 1 year: -40 °C ~ 35 °C (-40 °F ~ 95 °F)
Shell Material	Engineering plastic + engineering plastic alloy (PC + ABS)
Vehicle Interfaces	
CAN Bus	2 channels
Diagnostics	J1708 / OBD-II / J1939 supported (vehicle protocols)
RS-485	1 channel
RS-232	1 channel
Ignition Signal	1 channel
Digital Input	4 channels
Digital Output	3 channels (max 300 mA)
1-Wire	1 channel (supports 4 sensors)
Analog Input	1 channel + 1 channel

Category / Parameter	Specification
SIM Card	2 × FF, push-in slot
I/O PIN	26 PIN
LED Indicators	2 LED indicators (cellular status / GNSS status)

4. Software Specifications

Category / Parameter	Specification
Cloud Platforms & Connectivity	
Cloud Platforms	AWS IoT, Azure IoT, Aliyun IoT, Wialon, Traccar, GPSWox, WhiteLabel Tracking, ThingsBoard, Customer Platform
Transport Protocols	TCP / UDP / HTTP / MQTT
Vehicle Data & Transparent Transmission	
Vehicle Data	OBD-II, J1939, J1979 and J1708
Transparent Transmission	RS-232/RS-485 transparent; Modbus RTU to Modbus (data acquisition)
Event Alarm & Reporting	
Event Alarm	Collision detection, motion detection, overspeed, IO change, ignition signal detection
Reporting Support	SMS or FlexAPI over TCP/UDP/MQTT
ELD	Supported
BLE Forwarding	Forward vehicle data via BLE

Category / Parameter	Specification
Configuration Interface	
Configuration	RS-232 or Bluetooth; USB Type-C
Certificate (as listed)	
Certifications*	FCC / IC / PTCRB / AT&T / Verizon (model-dependent)

5. Ordering Information

Model Rule

Model code: VT320-<WMNN>

<WMNN>: Cellular Type & Module (LTE Cat M1 module variants)

Product Models

Model	Type / Region	Frequency / Bands (as listed)
VT320-FQ02	LTE Cat-M1 / Global	LTE-FDD: B1/B2/B3/B4/B5/B8/B12/B13/B14/B18/B19/B20/B25/B26/B27/B28/ B66/B85

Cable Accessories

Cable	Order Code	Specifications
26 PIN Cable	SCAB000370	P1 is 26 PIN female connected to VT320; P2 is open end requiring 9–48 V adaptor (engineering environments / indoor tests)
OBD 16 PIN Test Cable	SCAB000399	OBD 16 PIN interface test line; UL2464; wire length 1500 mm
J1939 6 PIN Test Cable	SCAB000409	J1939 6 PIN interface test line; UL2464; wire length 1500 mm
J1939 9 PIN Test Cable	SCAB000410	J1939 9 PIN interface test line; UL2464; wire length 1500 mm

6. Contact Us

- Website: [InHand Networks](https://www.inhandnetworks.com)
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7. Accessories

Accessory	Order Code	Specifications
26 PIN Cable	SCAB000370	The cable has P1 and P2 ends: P1 is 26PIN female, connected to VT320; P2 is open end, which requires a 9-48V adaptor. Suitable for engineering environments and indoor tests.
OBD 16PIN Test Cable	SCAB000399	OBD 16PIN interface test line; cable standard UL2464; wire length 1500 mm.
J1939 6PIN Test Cable	SCAB000409	J1939 6PIN interface test line; cable standard UL2464; wire length 1500 mm.
J1939 9PIN Test Cable	SCAB000410	J1939 9PIN interface test line; cable standard UL2464; wire length 1500 mm.
Quick terminal -3in3out	ECON060255	Rated voltage 600V, rated current 30A, 40*18.6*14.5 mm, flame retardant grade V0.
Power adapter 12V/2A (EU standard)	APWR000121	Switching power supply - 12V/2A - round connector - vertical (European standard) - DC line length 1.5 m, office test use.
Power adapter 12V/2A (US standard)	APWR000122	Switching power supply - 12V/2A - round connector - vertical (US standard) - DC line length 1.5 m, office test use.